MATH 55 - HW 3

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Section 2.3

12. Determine whether each of these functions from $\mathbb{Z}$ to $\mathbb{Z}$ is one-to-one. (justify your answers).

c) $f(n) = n^3$

If $f(a) = f(b)$, then:

$$a^3 = b^3$$

Taking cube roots (which is valid over the integers since the cube function is strictly increasing and one-to-one):

$$a = b$$

Therefore, $f(n) = n^3$ is **one-to-one**.

d) $f(n) = \lceil n/2 \rceil$

Evaluate at sample points:

$$f(1) = \lceil 1/2 \rceil = \lceil 0.5 \rceil = 1 = f(2) = \lceil 2/2 \rceil = \lceil 1 \rceil = 1$$

Since $f(1) = f(2)$ but $1 \neq 2$, the function is **not** one-to-one.

14. Determine whether $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is onto if (justify your answers).

b) $f(m, n) = m^2 - n^2$

Factor:

$$f(m, n) = (m - n)(m + n).$$

So the image is the set of all integers that can be written as the product of two integers of the **same parity** (since $m - n$ and $m + n$ are either both even or both odd).

- If both are even, their product is divisible by 4.

- If both are odd, their product is odd.

So the range is:

$$\{\text{all odd integers}\} \cup \{\text{multiples of } 4\}.$$

Missing values: integers of the form $4k + 2$
So **not onto**.

d) $f(m, n) = |m| - |n|$

Take arbitrary $z \in \mathbb{Z}$.

- If $z \geq 0$: choose $m = z, n = 0$. Then $f(m, n) = |z| - 0 = z$.
- If $z < 0$: say $z = -k$ with $k > 0$. Choose $m = 0, n = k$. Then $f(m, n) = 0 - |k| = -k = z$.

So indeed every integer is achievable. Thus f is **onto**.

34. Suppose that g is a function from A to B and f is a function from B to C. Prove each of these statements.

a) If $f \circ g$ is onto, then f is onto.

Take any $c \in C$. Since $f \circ g$ is onto, there exists $a \in A$ with

$$(f \circ g)(a) = c.$$

Let $b = g(a) \in B$. Then $f(b) = f(g(a)) = c$. Since c was arbitrary, every $c \in C$ has a preimage under f, so f is onto.

b) If $f \circ g$ is one-to-one, then g is one-to-one.

Suppose $g(a_1) = g(a_2)$ for some $a_1, a_2 \in A$. Apply f to both sides:

$$(f \circ g)(a_1) = f(g(a_1)) = f(g(a_2)) = (f \circ g)(a_2).$$

Because $f \circ g$ is one-to-one we must have $a_1 = a_2$. Hence g is one-to-one.

c) If $f \circ g$ is a bijection, then g is onto iff f is one-to-one.

Since $f \circ g$ is a bijection it is both onto and one-to-one.

($\Rightarrow$) Assume g is onto. To show f is one-to-one, let $b_1, b_2 \in B$ and suppose $f(b_1) = f(b_2)$. Because g is onto there exist $a_1, a_2 \in A$ with $g(a_1) = b_1$ and $g(a_2) = b_2$. Then

$$(f \circ g)(a_1) = f(g(a_1)) = f(b_1) = f(b_2) = f(g(a_2)) = (f \circ g)(a_2).$$

Since $f \circ g$ is one-to-one, $a_1 = a_2$; applying g gives $b_1 = g(a_1) = g(a_2) = b_2$. Thus f is one-to-one.

($\Leftarrow$) Assume f is one-to-one. To show g is onto, let $b \in B$. Consider $c = f(b) \in C$. Because $f \circ g$ is onto, there exists $a \in A$ with $(f \circ g)(a) = c$, i.e. $f(g(a)) = f(b)$. Since f is one-to-one it follows that $g(a) = b$. Hence every $b \in B$ has a preimage under g, so g is onto.

Combining both directions gives the desired equivalence.

40. Let $f(x) = ax+b$ and $g(x) = cx+d$, where a, b, c, and d are constants. Determine necessary and sufficient conditions on the constants a, b, c, and d so that $f \circ g = g \circ f$.

- $(f \circ g)(x) = f(g(x)) = f(cx + d) = a(cx + d) + b = acx + ad + b$
- $(g \circ f)(x) = g(f(x)) = g(ax + b) = c(ax + b) + d = acx + cb + d$

For $f \circ g = g \circ f$:

$$\begin{aligned} acx + ad + b &= acx + cb + d \\ ad + b &= cb + d \\ ad - d &= cb - b \end{aligned}$$

Factor (the necessary and sufficient condition for $f \circ g = g \circ f$ is):

$$d(a - 1) = b(c - 1)$$

50. Show that if x is a real number, then $\lceil x \rceil - \lfloor x \rfloor = 1$ if x is not an integer and $\lceil x \rceil - \lfloor x \rfloor = 0$ if x is an integer.

If x is an integer: suppose $x = n$ where $n \in \mathbb{Z}$.

- $\lfloor x \rfloor = n$
- $\lceil x \rceil = n$

So

$$\lceil x \rceil - \lfloor x \rfloor = n - n = 0.$$

If x is not an integer: suppose x lies strictly between two consecutive integers, say $n < x < n + 1$, where $n \in \mathbb{Z}$.

- $\lfloor x \rfloor = n$
- $\lceil x \rceil = n + 1$

So

$$\lceil x \rceil - \lfloor x \rfloor = (n + 1) - n = 1.$$

Conclusion:

$$\lceil x \rceil - \lfloor x \rfloor = \begin{cases} 0, & \text{if } x \text{ is an integer,} \\ 1, & \text{if } x \text{ is not an integer.} \end{cases}$$

Section 2.5

2. Determine whether each of these sets is finite, countably infinite, or uncountable. For those that are countably infinite, exhibit a one-to-one correspondence between the set of positive integers and that set.

c) Finite
d) Uncountable
e) Countably Infinite
f) Countably Infinite

10. Give an example of two uncountable sets A and B such that $A - B$ is (and explain).

a) $A = \mathbb{R},\ B = \mathbb{R} \setminus \{0\}$
b) $A = \mathbb{R},\ B = \mathbb{R} \setminus \mathbb{Q}$
c) $A = \mathbb{R},\ B = [0, 1]$

Additional problem

The bijection between $\mathbb{N}$ and $\mathbb{N} \times \mathbb{N}$ orders the pairs (i, j) along anti-diagonals where the sum $s = i + j$ is constant, starting with $s = 0$ and increasing. For each fixed s, the pairs are ordered by increasing j from 0 to s (equivalently, decreasing $i = s - j$).

To derive the formula for $f(i, j)$, first compute the number of pairs before the anti-diagonal with sum $s = i+j$. The anti-diagonal for each k from 0 to $s-1$ contains exactly $k + 1$ pairs, so the total is $\sum_{k=0}^{s-1}(k + 1) = \sum_{m=1}^{s} m = \frac{s(s+1)}{2}$.

Within the anti-diagonal for sum s, the pair (i, j) is the $(j + 1)$-th element (since j starts at 0), so add j to the previous total.

Thus, $f(i, j) = \frac{(i+j)(i+j+1)}{2} + j$, which matches the positions in the diagram.